

CAMAT_v2

Cloud-Based Tools for Symbolic Music Analysis

Egor Polyakov • Martin Pfeleiderer • University of Music Franz Liszt Weimar

The project

CAMAT (*Computer-Assisted Music Analysis Toolbox*) is an open-source Python toolbox for digital editions and computational analysis of music scores encoded in **MEI** (*Music Encoding Initiative*).

Four project objectives:

- Edit music in the MEI data format
- Parse music scores into analytical representations
- Analyse music with statistics and pattern search
- Provide tutorials, analytical case studies and documentation

CAMAT aims to make **Jupyter notebooks** and MEI easier to use in musicological and music-theoretical research and education, with integrated tools for score representation, visualisation and music analysis.

Editions and corpora

Our editions start from page-level **OMR** (*Optical Music Recognition*) supplied through the **Bavarian State Library (BSB)** and *musiconn.scoresearch*. Unfortunately, these machine-generated encodings contain many **recognition errors** and require editorial correction.

- **Denkmäler deutscher Tonkunst I/11: Buxtehude's instrumental works.** Fourteen sonatas and three appendix works (ed. Carl Stiehl, 1903)
- **Denkmäler deutscher Tonkunst I/29–30: Instrumentalkonzerte deutscher Meister** (ed. Arnold Schering, 1907). Pisendel, Hasse, C. P. E. Bach, Telemann, Graupner, Stölzel and Hurlbusch
- **Early twentieth-century corpus:** selected works by Anton Webern, Arnold Schoenberg, Charles Ives and Béla Bartók

01 Edit music with MEI

Prepare the sources: Use page-level OMR of facsimiles or convert digital scores in various formats into MEI (Fig. 1).

Assemble an edition: Join page-level OMR files into complete works; add metadata and link encoded measures to their regions in the facsimiles through IIIF (International Image Interoperability Framework).

Correct and review: Correct recognition errors in online MEI editor **mei-friend**; review edits with CAMAT's facsimile viewer.

Check and repeat: Use automated reports to flag inconsistent durations and broken identifiers or facsimile links, then correct and re-check the MEI file.

02 Parse CAMAT representations

Extract tables: Parse MEI with Verovio into note data (`df_pitch`) and non-note events (`df_events`), retaining pitch, timing, voice, measure and source identifiers.

Select and visualise: Filter voices or passages. CAMAT supports two plotting backends: Matplotlib provides static plots; Bokeh adds hover information, panning and zooming. Verovio renders the notation.

Annotate a selection: Attach an analytical label or comment to notes or a measure/time range in MEI; view the text and linked highlights in **mei-friend** or CAMAT's viewers.

Build binary representations: Map piano-roll data to occupied pitch/time cells for pattern search, retaining metadata that links cells to source notes.

03 Analyse music

Statistics: Compare distributions of pitches, pitch classes, durations and metrical positions across selected voices and passages; examine melodic intervals and pitch transitions within individual voices.

Pattern search: The convolution-based search slides a pattern across the pitch/time grid and scores each position. Search settings support transposed and rhythmically scaled patterns.

1. **Define a query:** Extract or draw binary kernels as pitch/duration masks for motifs, chord progressions or textures (Fig. 2).
2. **Find and locate matches:** Rank candidate passages by similarity or containment; inspect their positions in the pitch/time grid (Fig. 3).
3. **Interpret the results:** Project matches onto piano-rolls and the original MEI notation. Retained metadata identifies the contributing notes (Fig. 4).

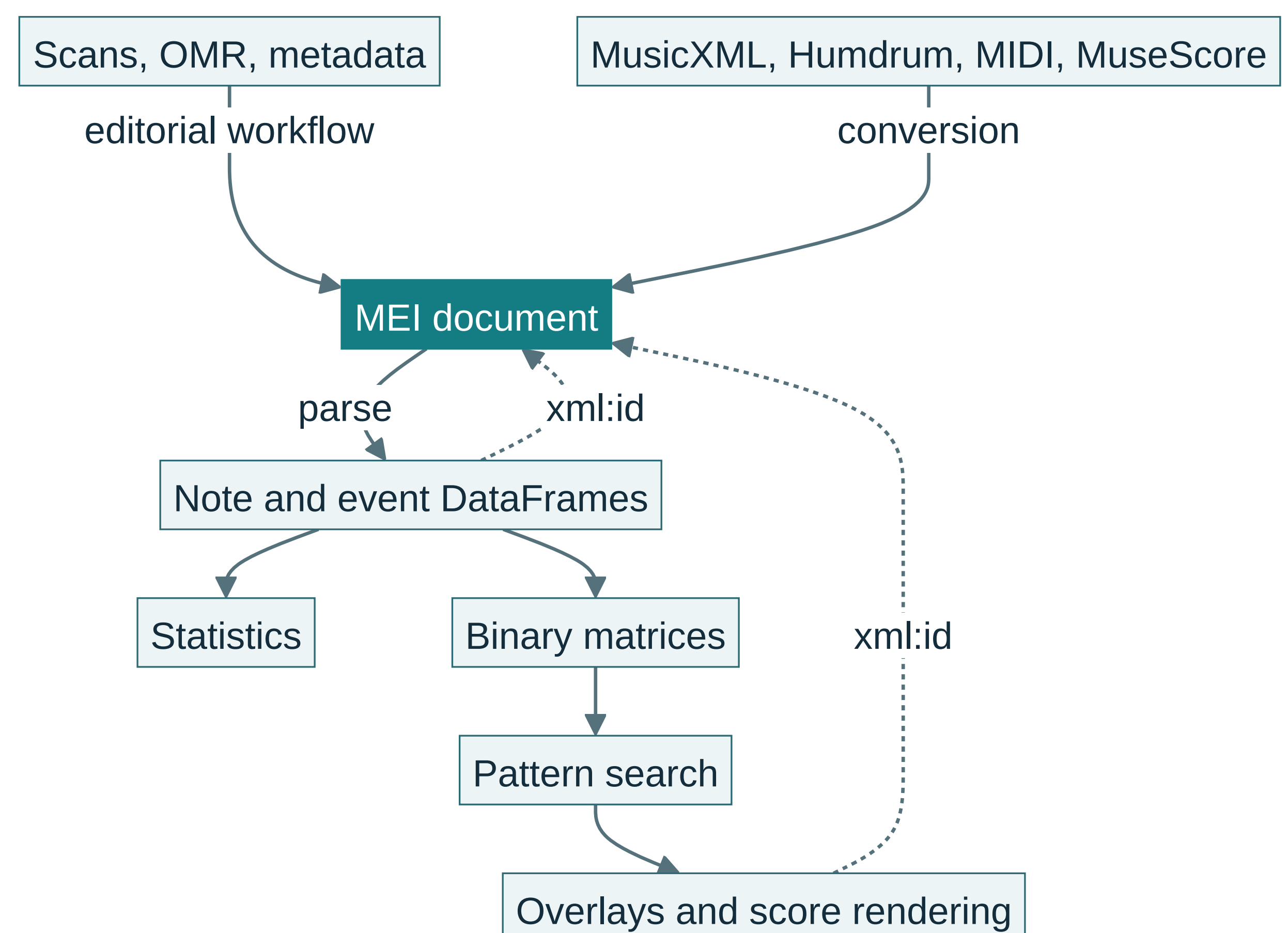


Fig. 1. Editing, parsing and analysing share the same MEI document. Identifiers and retained metadata connect results to the original notation.

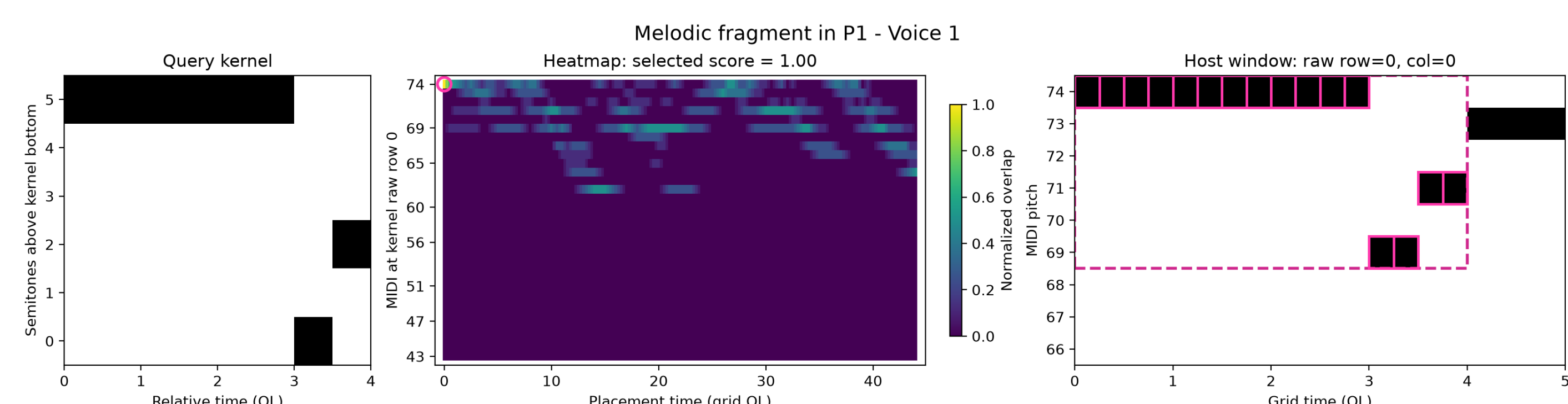


Fig. 2. Search for a melodic fragment from Johann Sebastian Bach's *Ein feste Burg*: query kernel (left), score map (centre) and selected matching window (right).

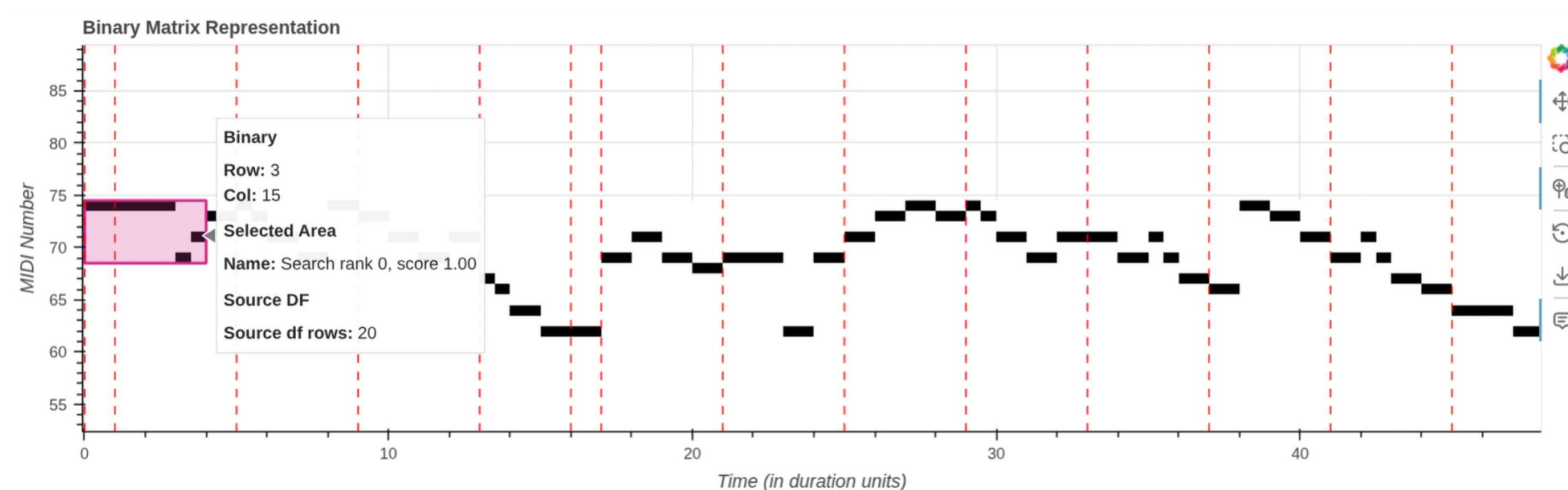


Fig. 3. Locate the selected match in the binary pitch/time grid. Metadata links the highlighted cells to the source notes.



Fig. 4. Inspect the contributing notes in the original MEI score rendered with Verovio. Magenta highlights show the matched musical cell.

04 Tutorials, case studies and documentation

Tutorials and case studies: A growing collection of Jupyter notebooks connects technical explanations with case studies and tutorials in music analysis, including motif searches and chord-progression analysis.

Cloud access to Jupyter notebooks: Our collaborator **Jupyter4NFDI** (Jülich Supercomputing Centre, JSC) lets researchers, lecturers and students use Jupyter notebooks in the browser, without installing a local Python environment. CAMAT provides dedicated workflows for cloud deployment.

Documentation: The GitHub repository and documentation site provide a wiki, Jupyter notebook tutorials, an API reference and cloud setup guidance.

Scan the QR code below to get started!

Funded by the German Research Foundation (DFG): *A Comprehensive Cloud-Based Toolbox for Sheet Music Analysis*. LIS / E-Research Technologies, PF 669/18-1.

Principal investigator: Martin Pfeleiderer.

Researcher: Egor Polyakov.

Research assistant: Pia Steuck.

Edition contributors: Francy Häring, Jan-Raphael Kemnitzer, Jingyu Tu, Lea Meret Beck, Ragna Conrads and You Zhou.

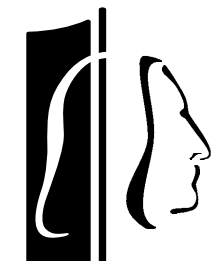
Code: https://github.com/egorpol/camat_v2

Docs: <https://camat-v2.readthedocs.io/en/stable/>

Editions: https://github.com/egorpol/DdT_1_vol_11

https://github.com/egorpol/DdT_1_vol_29_30

Cloud: <https://nfdi-jupyter.de/>



Hochschule für Musik
FRANZ LISZT Weimar

Funded by



Deutsche
Forschungsgemeinschaft
German Research Foundation

